# E-COMMERCE SALES TREND ANALYSIS

Project 3 – Business Brief

## 1. Context

This analysis uses a synthetic e-commerce dataset of 1,500 fictional orders from 2025. It examines monthly revenue, product-category performance, and customer spending to identify patterns and suggest practical business actions.

## 2. Key Findings

| Area | Finding |
|---|---|
| Monthly revenue | April had the highest revenue (164,116.94); June had the lowest (122,242.94). |
| Category | Fashion generated the highest revenue (565,121.54) and the most orders (505). |
| Customers | The top 10% of customers (29 of 295) contributed 19.50% of total revenue. |

## 3. Recommendations

| Action | Suggested step |
|---|---|
| Strengthen Fashion | Fashion leads in revenue and order count. Maintain stock of popular items and test targeted promotions. |
| Investigate June | June was the lowest-revenue month. Review demand, product availability, and promotional activity around this period. |
| Encourage repeat purchases | Pilot a loyalty or repeat-purchase campaign and measure its effect on repeat orders and revenue. |

## 4. Next Steps

Compare results across additional sales periods, investigate the reasons for monthly changes, and monitor category revenue, average order value, and repeat-purchase rates after implementing any actions.

**Data source and limitation:** The dataset was synthetically generated with Python and NumPy for educational purposes and contains no real customer or transaction records. This project analyses sales trends and customer spending; it does not build a churn prediction model.